Risk-based Portfolio Construction

Motivation for Risk-based Optimization

- Common portfolio optimisations are based on both Risk and Return assumption.
- Return is hard to forecast, and small estimation error will lead to huge deviations from optimal results, if no additional constraints
- One common characteristic across all of risk-based portfolio optimizations is that the only input required to determine the portfolio compositions is a model of risk, which is typically measured by the covariance matrix, while explicit modeling of expected returns is not required.
- Traditional asset allocation method overlooks the effect of “Hierarchical Structure”
- Are we able to combine the machine learning method with portfolio optimisation and take the hierarchy of assets into consideration?

Status Quo of Portfolio Construction

Risk Based
(View Irrelevant)

Highlight the portfolio diversification

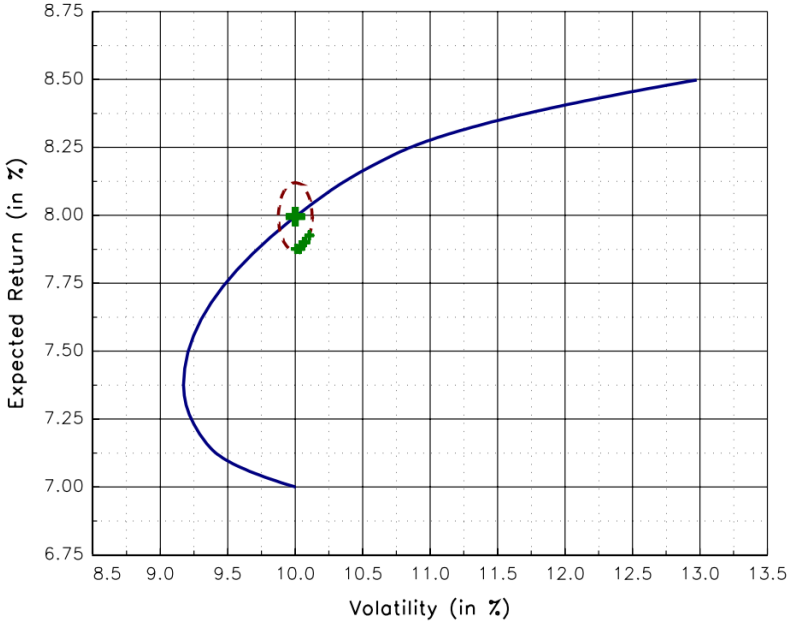
MIN Variance

Risk Parity

Clustering-Based

Equal Weight

The simplest strategy without taking risk and return information



Return Based
(View Driven)

Objective is to maximise the return

Mean Variance
(Based on Single CML)

Black-Litterman
(Including view in CML)

Robust Opt
(Including range of CML)

Method 1: Inverse Volatility

- The first allocation technique we explore is Inverse Volatility (InvVol), also called naïve risk parity or volatility parity. InvVol is easy to implement, in that portfolios are weighted inversely to their volatility. As such, more volatile assets are relatively down weighted.
- Mathematically, this can be expressed as follows:

$$\omega_{i,t} = \frac{1/\sigma_{i,t}}{\sum_{n} 1/\sigma_{i,t}}$$

- The main drawback of InvVol is that it does not take into account the correlation between assets. An asset may be unnecessarily penalized (i.e. down weighted) simply because it's relatively more volatile, while it may provide more diversification benefits should correlations also be considered.

Method 2: Minimum Variance

- The MinVar method aims to weight portfolios such that the overall portfolio risk is minimized without taking any particular view on expected returns. This can be implemented via the following optimization algorithm:

$$\arg\min_{\omega} \frac{1}{2} \omega_t' \Sigma_t \omega_t$$

- MinVar portfolios are sensitive to our risk estimates and tend to be concentrated in a few assets.

Method 3: Risk Parity (Equal Risk Contribution)

Let $\sigma(x) = \sqrt{x^\top \Sigma x}$ be the risk of the portfolio x. The Euler decomposition gives us:

$$\sigma(x) = \sum_{i=1}^{n} \sigma_i(x) = \sum_{i=1}^{n} x_i \times \frac{\partial \sigma(x)}{\partial x_i}.$$

with $\partial_{x_i} \sigma(x)$ the marginal risk contribution and $\sigma_i(x) = x_i \times \partial_{x_i} \sigma(x)$ the risk contribution of the ith asset.

Starting from the definition of the risk contribution $\sigma_i(x)$, the idea of the ERC strategy is to find a risk-balanced portfolio such that the risk contribution is the same for all assets of the portfolio:

$$\sigma_i(x) = \sigma_j(x)$$

Method 3: Risk Parity (Equal Risk Contribution)

- The conditions for three risk-based portfolios:
  - ew

$$x_i = x_j$$

  - mv

$$\frac{\partial \sigma(x)}{\partial x_i} = \frac{\partial \sigma(x)}{\partial x_j}$$

  - erc

$$x_i \frac{\partial \sigma(x)}{\partial x_i} = x_j \frac{\partial \sigma(x)}{\partial x_j}$$

- The ERC portfolio may be viewed as a portfolio “between” the 1/n portfolio and the minimum variance portfolio.

Method 3: Risk Parity (Equal Risk Contribution)

- The general solution to ERC optimization is equivalent to solving the following non-linear optimisation problem:

$$\begin{aligned} x^{\star} &= \arg\min f(x) \\ &\text{u.c. } \mathbf{1}^{\top} x = 1 \text{ and } \mathbf{0} \leq x \leq \mathbf{1} \end{aligned}$$

$$f(x) = n \sum_{i=1}^{n} x_i^2 (\Sigma x)_i^2 - \sum_{i,j=1}^{n} x_i x_j (\Sigma x)_i (\Sigma x)_j$$

- Special case 1: if the correlations are the same, the solution is

$$x_i = \frac{\sigma_i^{-1}}{\sum_{j=1}^{n} \sigma_j^{-1}}$$

- Special case 2: if the volatilities are the same, the solution is

$$x_i \propto \left(\sum_{k=1}^{n} x_k \rho_{ik} \right)^{-1}$$

Method 3: Risk Parity (Equal Risk Contribution)

- We consider a universe of 4 risky assets. Volatilities are respectively 10%, 20%, 30% and 40%.
- If the correlations are constant (assume 0.5)

The $1/n$ portfolio

$\sigma(x) = 20.2\%$	x_i	$\partial_{x_i}\sigma(x)$	$x_i \times \partial_{x_i}\sigma(x)$	$c_i(x)$
1	25%	0.068	0.017	8.5%
2	25%	0.149	0.037	18.5%
3	25%	0.242	0.060	30.0%
4	25%	0.347	0.087	43.1%

Method 3: Risk Parity (Equal Risk Contribution)

The minimum variance portfolio

$\sigma(x) = 10.0\%$	x_i	$\partial_{x_i}\sigma(x)$	$x_i \times \partial_{x_i}\sigma(x)$	$c_i(x)$
1	100%	0.100	0.100	100%
2	0%	0.100	0.000	0%
3	0%	0.150	0.000	0%
4	0%	0.200	0.000	0%

The ERC portfolio

$\sigma(x) = 15.2\%$	x_i	$\partial_{x_i}\sigma(x)$	$x_i \times \partial_{x_i}\sigma(x)$	$c_i(x)$
1	48%	0.079	0.038	25%
2	24%	0.158	0.038	25%
3	16%	0.237	0.038	25%
4	12%	0.316	0.038	25%

Method 3: Risk Parity (Equal Risk Contribution)

The case of a general correlation matrix

$$\rho = \begin{pmatrix} 1.00 & & & \\ 0.80 & 1.00 & & \\ 0.00 & 0.00 & 1.00 & \\ 0.00 & 0.00 & -0.50 & 1.00 \end{pmatrix}$$

The $1/n$ portfolio

$\sigma(x) = 11.5\%$	x_i	$\partial_{x_i}\sigma(x)$	$x_i \times \partial_{x_i}\sigma(x)$	$c_i(x)$
1	25%	0.056	0.014	12.3%
2	25%	0.122	0.030	26.4%
3	25%	0.065	0.016	14.1%
4	25%	0.217	0.054	47.2%

Method 3: Risk Parity (Equal Risk Contribution)

The minimum variance portfolio

$\sigma(x) = 8.6\%$	x_i	$\partial_{x_i}\sigma(x)$	$x_i \times \partial_{x_i}\sigma(x)$	$c_i(x)$
1	74.5%	0.086	0.064	74.5%
2	0%	0.138	0.000	0%
3	15.2%	0.086	0.013	15.2%
4	10.3%	0.086	0.009	10.3%

The ERC portfolio

$\sigma(x) = 10.3\%$	x_i	$\partial_{x_i}\sigma(x)$	$x_i \times \partial_{x_i}\sigma(x)$	$c_i(x)$
1	38.4%	0.067	0.026	25%
2	19.2%	0.134	0.026	25%
3	24.3%	0.106	0.026	25%
4	18.2%	0.141	0.026	25%

Risk Parity in Asset Allocation

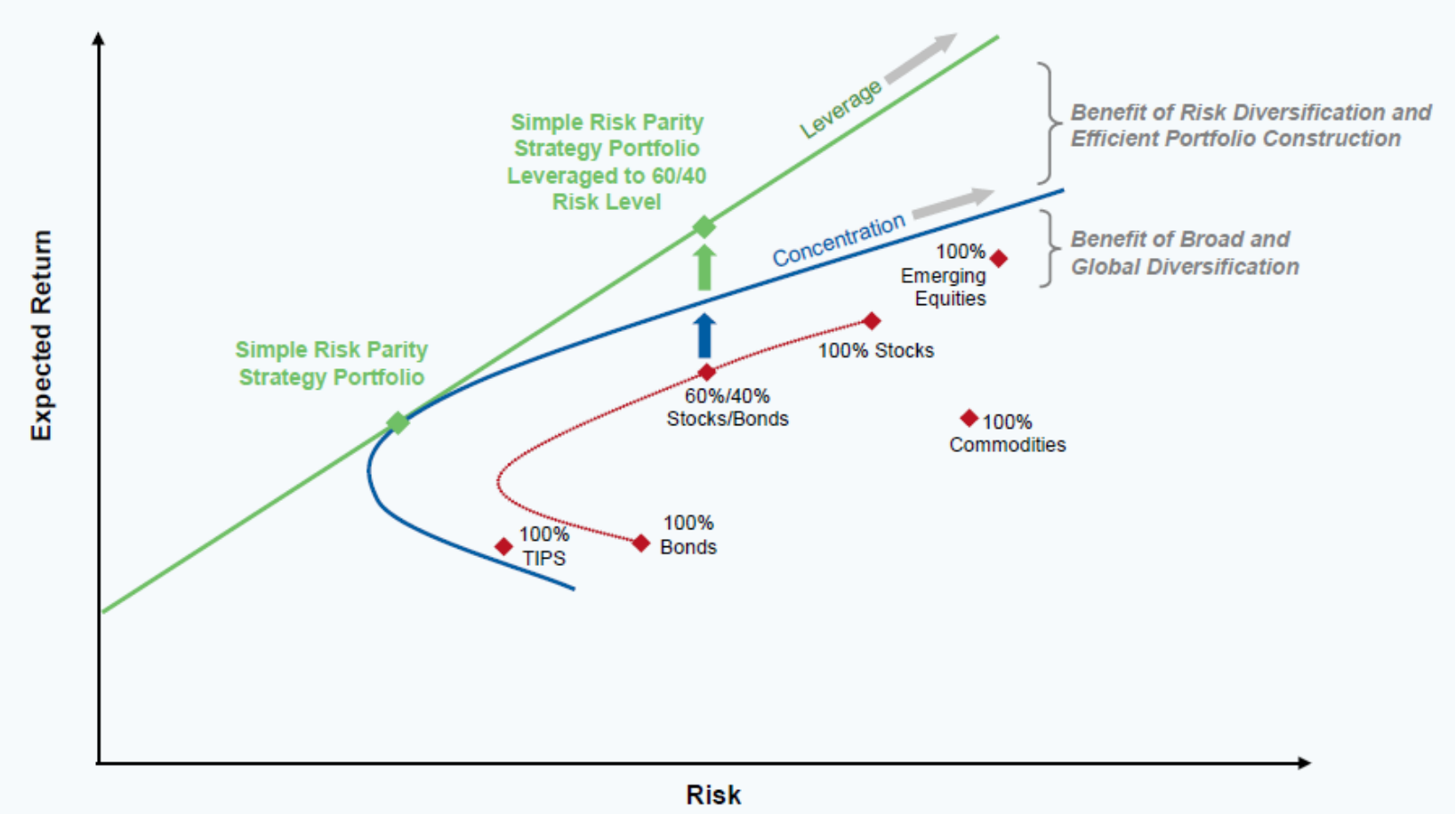
- Challenge 1: Some critics of risk parity make the claim that using volatility to determine how to allocate assets is not an appropriate measure of risk.

- Challenge 2: While there may be index providers and others who use historical volatility as a quick and transparent way to quantify the risk of an asset, seasoned risk parity managers generally utilize more sophisticated and robust methodologies to try and capture multiple dimensions of risk.

- Challenge 3: Relative capital allocation in a risk parity strategy is not static, but instead very much on the underlying risk regime at a particular point in time.

- Challenge 4: Because risk parity managers prefer low risk assets which tend to carry relatively low returns compared to higher risk assets, leverage may indeed be necessary to bring the expected return of the portfolio in line with an investor's required return objective.

Risk Parity in Asset Allocation



Method 4: Maximum Diversification

- This allocation strategy attempts to create portfolios that are more diversified by maximizing the distance between the weighted average volatility of each underlying portfolio and the overall portfolio volatility. This can be shown in the following equation:

$$\arg\max_{\omega} \frac{\sum_{n} \omega_{i,t}\sigma_{i,t}}{\omega_t' \sum_t \omega_t}$$

- The numerator is simply the weighted sum of the underlying asset volatilities. The denominator is the total portfolio volatility which takes into account the correlation between the underlying assets. The difference between the two is essentially the correlation terms.

Method 4: Maximum Diversification

- To maximize the overall ratio, the denominator containing the correlations must be minimized. This allocation strategy attempts to select assets that minimize the correlation between the underlying assets and hence “maximize diversification” as the name suggests.

- Numerically, the MaxDiversification portfolio can be solved easily by minimizing $\psi' P \psi$, where P is the correlation matrix. Therefore, the MaxDiversification optimization is almost the same as MinVar – the difference is to replace the covariance matrix with the correlation matrix. The final weights are then retrieved by rescaling the intermediate weight vector (optimized using the correlation matrix) with the standard deviations of the asset returns.

Comparison of Risk-based Optimizations

	Objective	Parameters	Optimality Conditions	Portfolio Characteristics
Equal Weight	Equalize capital weights	Number of assets	Similar returns and covariances	Minimal concentration, highly sensitive to investment universe
Inverse Volatility	Weights proportional to inverse of volatility	Asset volatilities	Similar Sharpe ratios and homogeneous correlations	Low concentration, highly sensitive to investment universe
Capitalization Weight	Weights proportional to market cap	Covariance with market portfolio	Similar Treynor ratios	Low concentration, sensitive to investment universe
Equal Risk Contribution	Equalize risk contributions	Covariance matrix	Similar *marginal* Sharpe ratios	Low concentration, moderately sensitive to covariances, moderately sensitive to investment universe
Maximum Diversification	Maximize portfolio diversification	Covariance matrix	Similar Sharpe ratios	Sometimes concentrated, sensitive to correlations, moderately sensitive to volatilities, robust to investment universe
Minimum Variance	Minimize portfolio volatily	Covariance matrix	Similar returns	Sometimes concentrated, sensitive to covariances, robust to investment universe
Mean-Variance	Maximize return subject to risk tolerance	Expected returns + Covariance matrix	NA	Sometimes concentrated, sensitive to return and covariance estimates, robust to investment universe

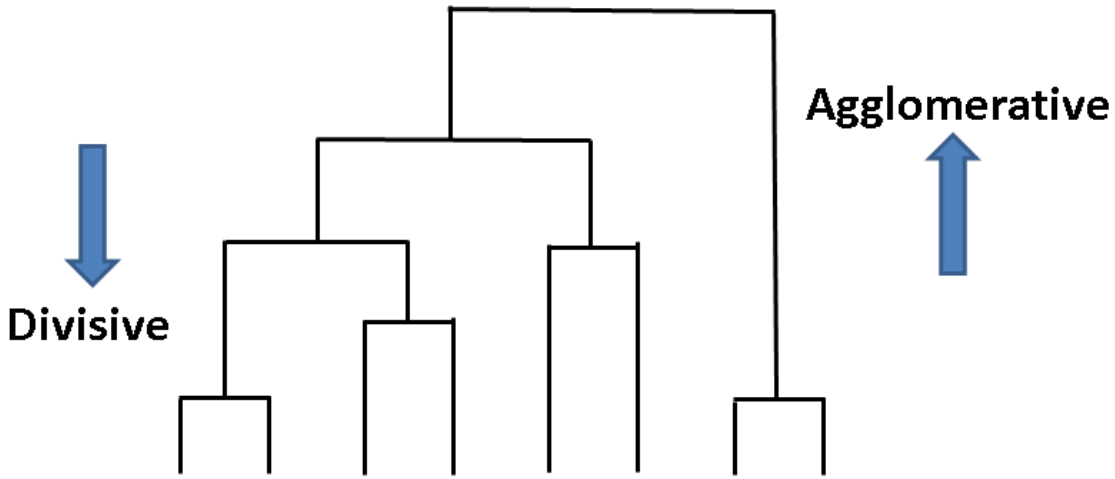
Combine ML with Risk Parity

- Clustering analysis has been widely used in market research, pattern recognition and image processing, before introduced to portfolio construction

- Hierarchical Risk Parity approach solved the concerns of quadratic optimizers in general: Instability, concentration and underperformance and based on graph theory and machine learning

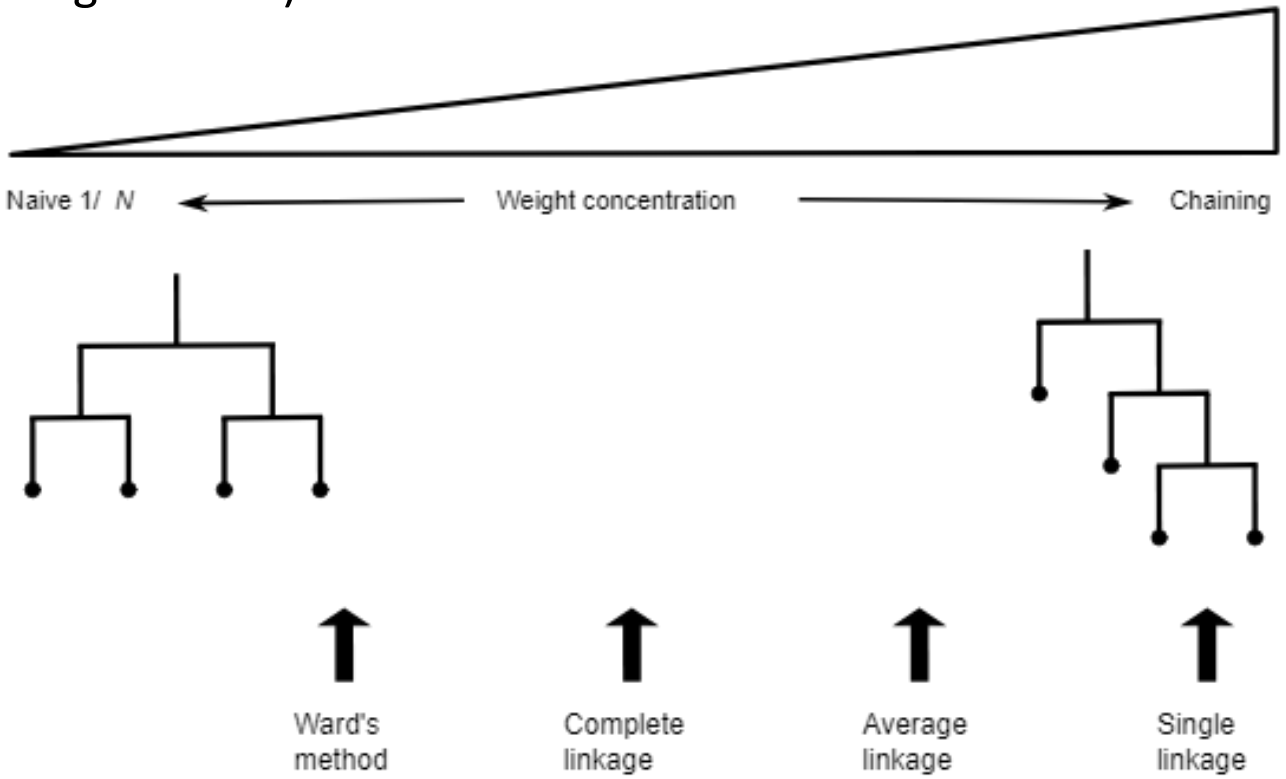
Combine ML with Risk Parity

- Hierarchical Clustering:
- A method of cluster analysis which seeks to build a hierarchy of clusters (Agglomerative vs. Divisive)
- Distance measure: $d = \sqrt{(2 * (1 - correlation))}$



Combine ML with Risk Parity

Clustering linkage method: recommended methods are single linkage (for high-risk investors willing to take high stakes in single assets) and Ward's method (for risk-averse investors with less concentration on single assets)



Hierarchical Clustering Portfolio Construction

Build Hierarchical Clustering Tree based on dissimilarity measure

- Correlation based
- Tail dependence coefficients based

Decide the optimal number of clusters

- Each end node is one cluster
- Selected optimal number visionally or by statistics

Decide how to make allocation within-cluster and across-clusters

- Equal Weighting: 1/N
- Min-variance
- Equal risk contribution
- Inverse Volatility
- …

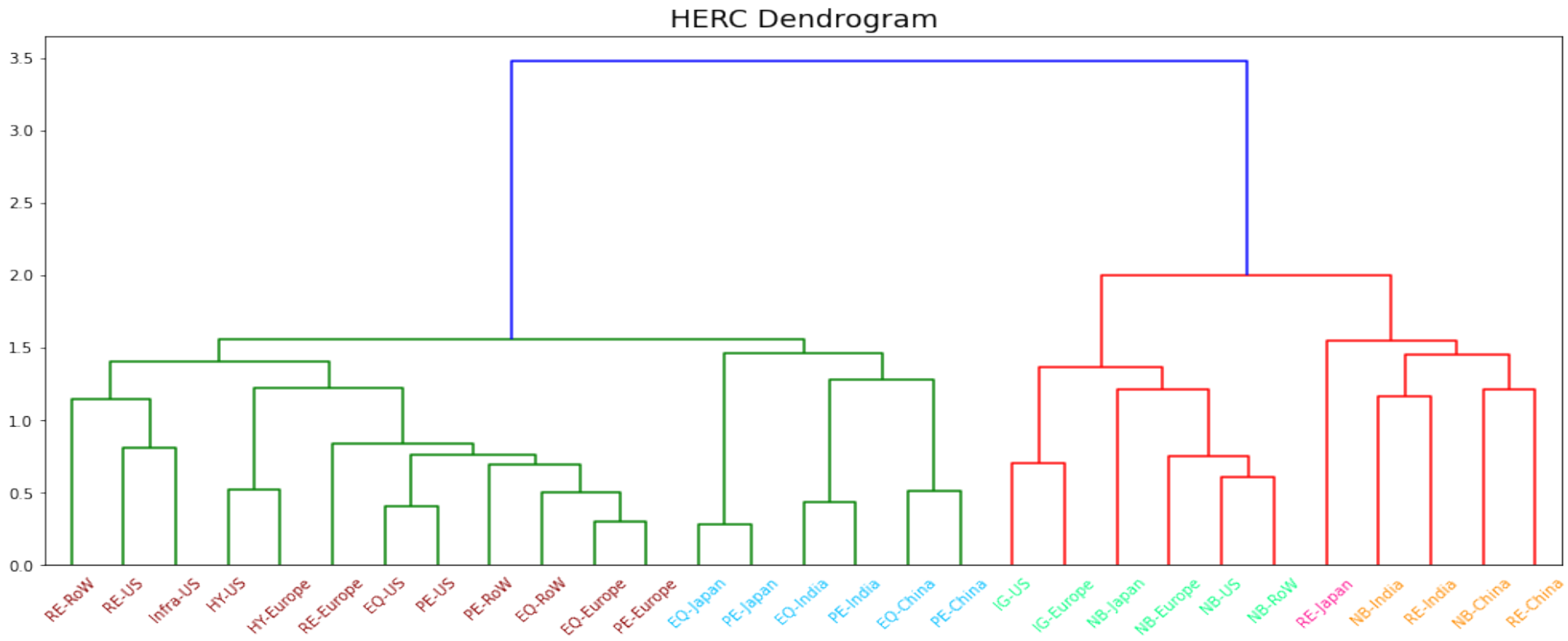
- Advantages :
  - The inversion of covariance matrix is not required, therefore the solutions will be stable.
  - The strategy provides autonomy in choosing the allocation methods within clusters or among clusters.
  - The Hierarchical effect is taken into consideration when allocate both risk and $value

Hierarchical Clustering Tree - Example

- Assets Classes : NB, EQ, PE, RE, INFRA, IG, HY
- Countries/Regions : US, Europe, Japan, China, India, ROW

Assets	Nominal Bond						Equity						Private Equity						Real Estate						Infra	IG		HY	
Country	US	Eur	Japan	China	India	RoW	US	Eur	Japan	China	India	RoW	US	Eur	Japan	China	India	RoW	US	Eur	Japan	China	India	RoW	US	US	Eur	US	Euro
Average Correlation (with all assets)	-0.05	0.00	-0.01	-0.02	0.04	0.01	0.38	0.38	0.29	0.32	0.35	0.42	0.39	0.38	0.28	0.32	0.34	0.40	0.26	0.30	0.00	0.01	0.16	0.23	0.34	0.28	0.26	0.37	0.34
Average Correlation (with equities)	-0.29	-0.21	-0.17	-0.10	-0.01	-0.21	0.69	0.69	0.57	0.56	0.54	0.74	0.73	0.72	0.60	0.57	0.58	0.70	0.45	0.60	0.00	-0.02	0.23	0.41	0.59	0.23	0.26	0.70	0.56
Volatility	0.05	0.05	0.02	0.04	0.07	0.05	0.15	0.18	0.18	0.26	0.24	0.13	0.18	0.18	0.17	0.28	0.27	0.13	0.11	0.09	0.05	0.09	0.27	0.05	0.12	0.06	0.04	0.09	0.12

Hierarchical Clustering Tree - Example

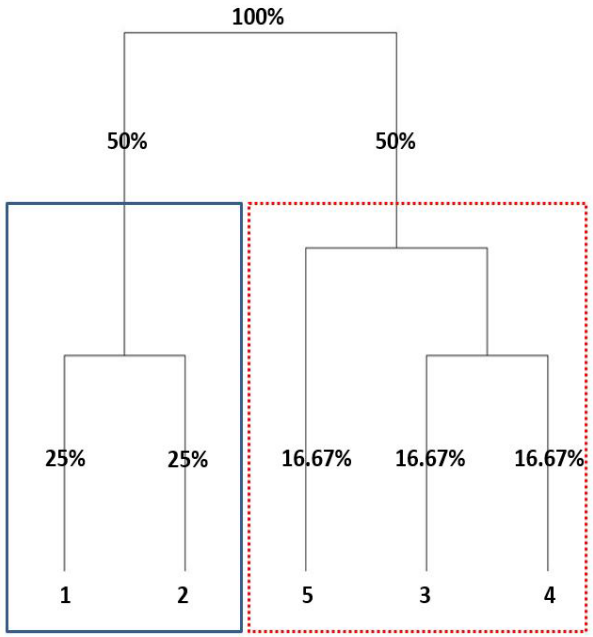


- More intuitive and homogenous clustering. Bond-like assets are close to each other, while different to Equity-like assets. Asia country effects also impact the clustering results

- However, it is challenging to handle the asset/country specific or granular constraints

Hierarchical Equal Weight Allocation

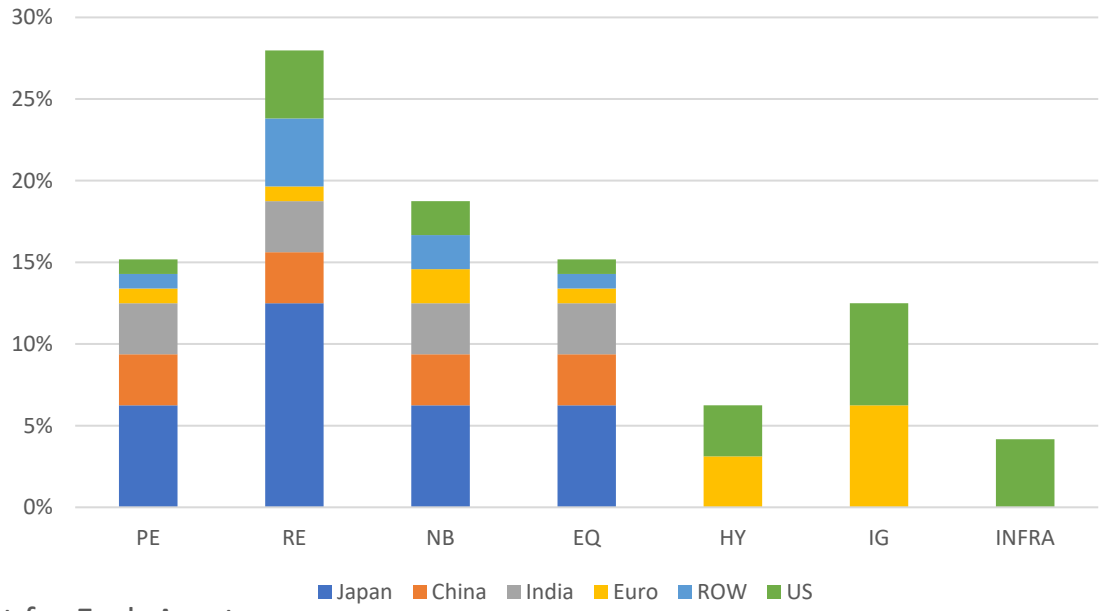
- Equal weights are allocated for merged clusters and for the assets within clusters.
- Simple example with 2 clusters and 5 assets is illustrated below
- Pros: simple implementation and the hierarchical structure is embedded in the strategy, that results in less concentrated portfolio
- Cons: the volatility of each assets and clusters are missed in the strategy, and it may lead to less risk diversified portfolio



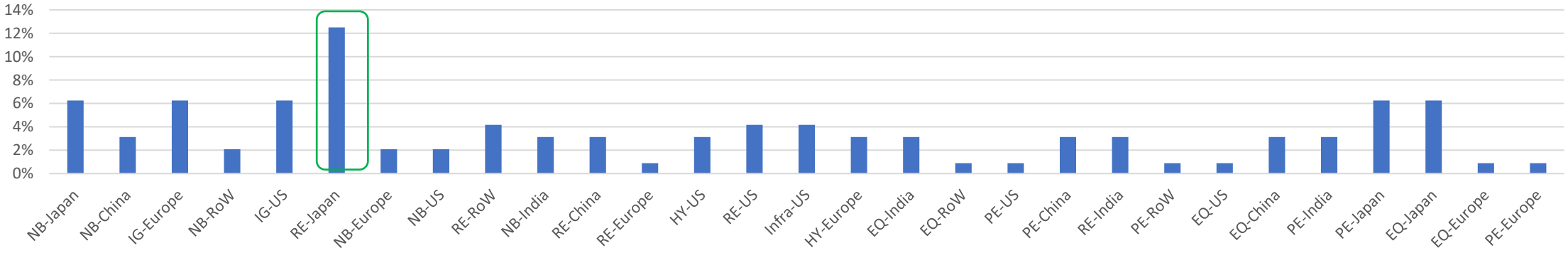
Hierarchical Equal Weight Allocation

Country Weights	
Japan	31.3%
China	12.5%
India	12.5%
Euro	14.1%
ROW	8.0%
US	21.6%

Asset Weights	
PE	15.2%
RE	28.0%
NB	18.8%
EQ	15.2%
HY	3%
IG	12.5%
INFRA	4.2%

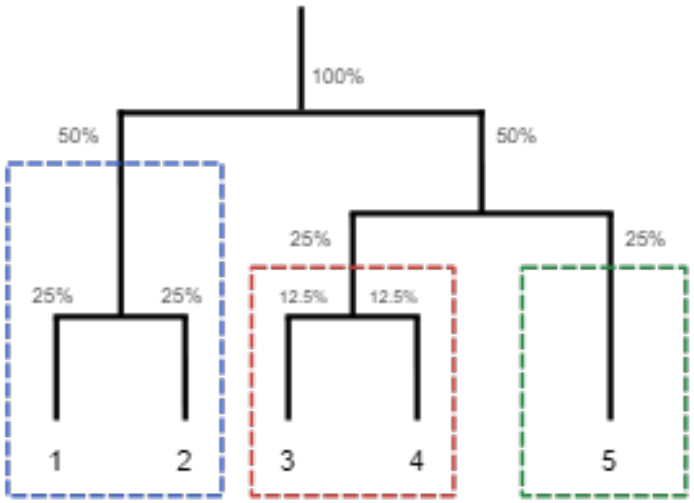


The weights concentrated in Japan as its assets is less similar to others

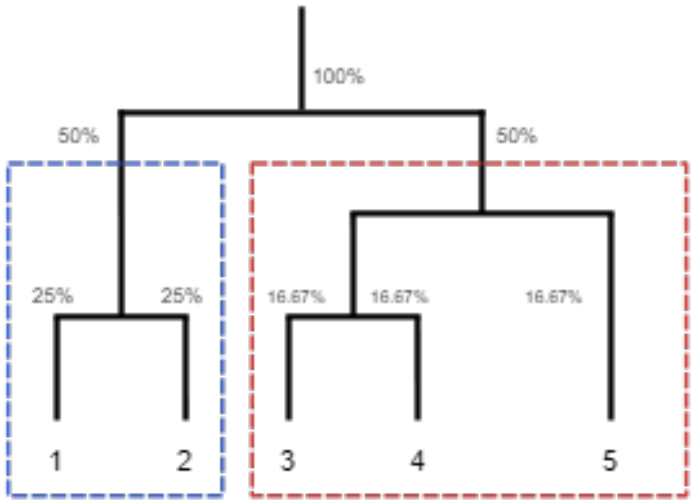


Hierarchical Equal Weight Allocation

- Early stopping in the hierarchical clustering to automatically select the appropriate number of clusters based on the gap statistic, a technique already used in hierarchical clustering asset allocation, as opposed to clustering all the way down to single assets as typically done in hierarchical clustering.



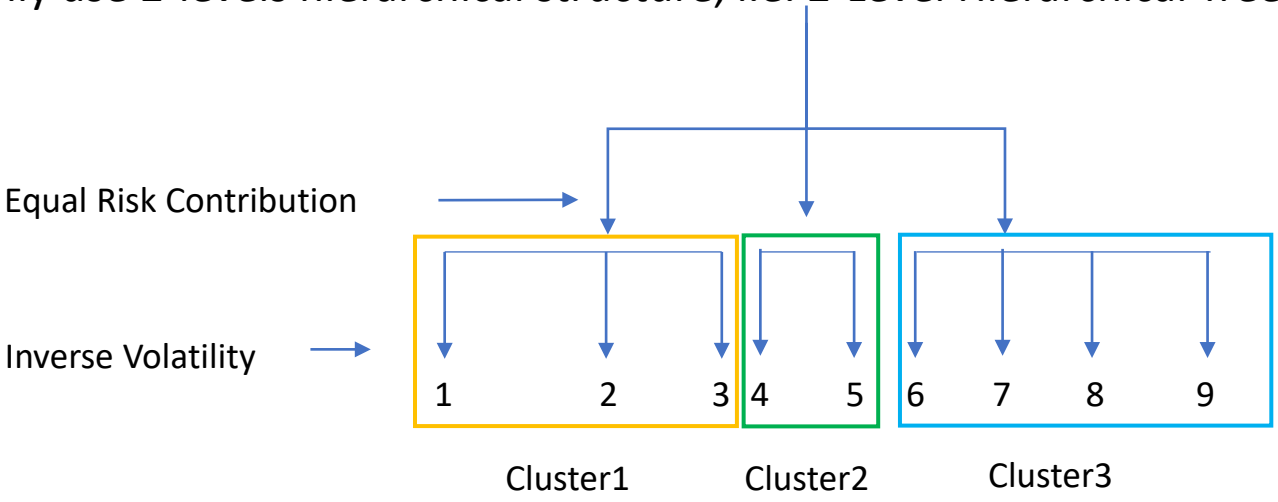
Three clusters



Two clusters

Two-Level Hierarchical Risk Parity

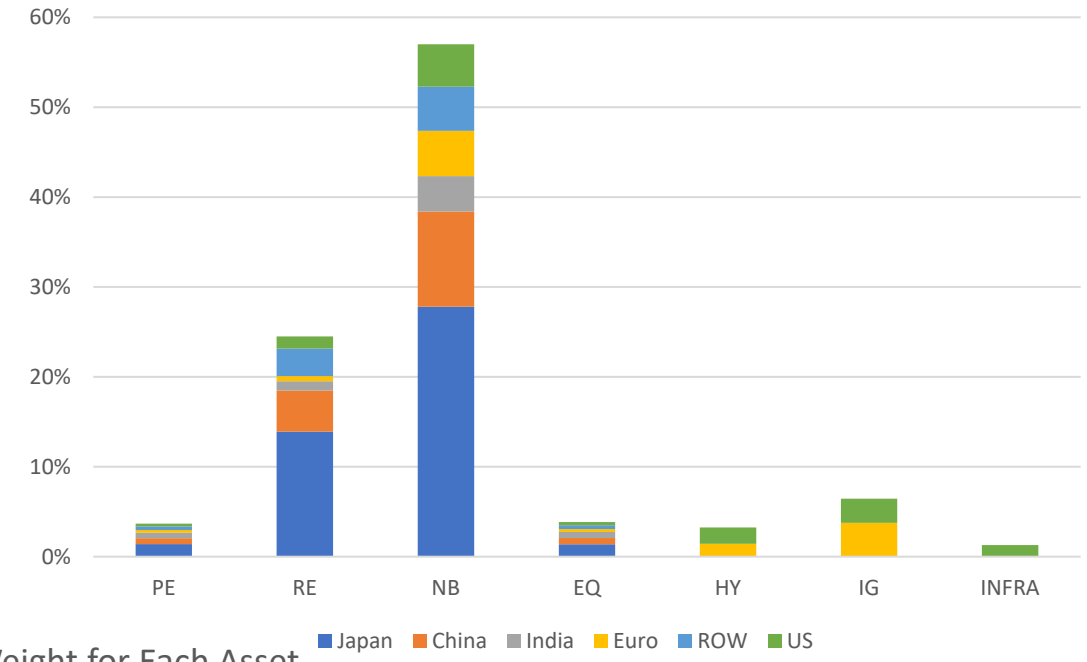
- This strategy combines inverse volatility and equal risk contribution methods, and all clusters are merged in one step instead of following multi-level hierarchical ordering
- Assets weights within each cluster are decided by Inverse Volatility, as the assets within same cluster are usually highly correlated to each other
- Cross-clusters weights are calculated based on Equal Risk Contribution
- Pros: Assets and clusters volatility are considered, still able to setup country and asset constraints at cluster level
- Cons: Only use 2-levels hierarchical structure, i.e. 2-Level Hierarchical Tree



Two-Level Hierarchical Risk Parity

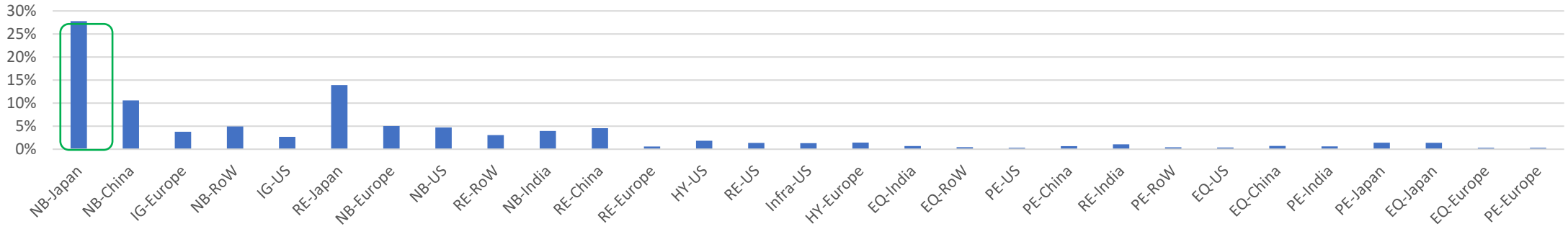
Country Weights	
Japan	44.5%
China	16.5%
India	6.3%
Euro	11.4%
ROW	8.8%
US	12.5%

Asset Weights	
PE	3.7%
RE	24.5%
NB	57.0%
EQ	3.9%
HY	3.3%
IG	6.4%
INFRA	1.3%



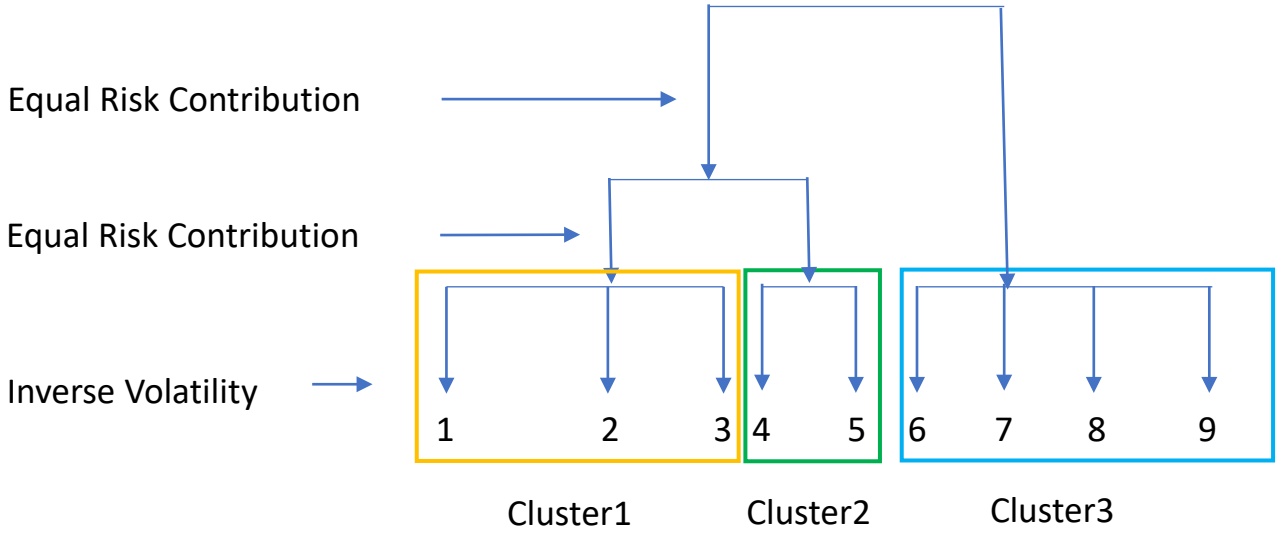
Weight for Each Asset

The weights concentrated in NB as its volatility is much lower than EQ



Multi-level Hierarchical Risk Parity

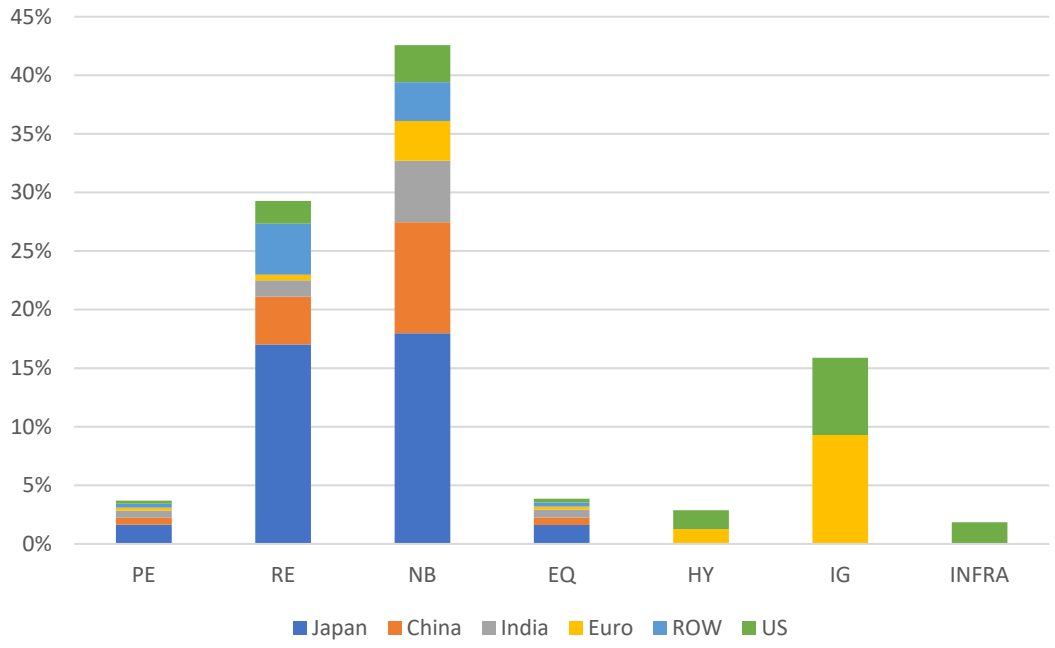
- This strategy still combines inverse volatility and equal risk contribution methods, but clusters are merged with hierarchical ordering
- Assets weights within each cluster are decided by Inverse Volatility
- Cluster weights are assigned based on Equal Risk Contribution at each step
- Pros: Allocation is based on both assets and clusters volatility, and whole hierarchical structure is considered.
- Cons: Not possible to setup top-down constraints (by assets or countries) during the allocation process



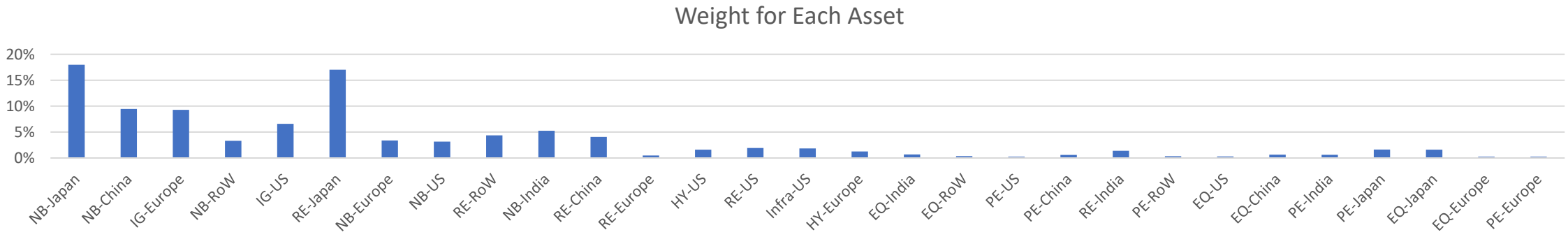
Multi-level Hierarchical Risk Parity

Country Weights	
Japan	38.3%
China	14.8%
India	7.9%
Euro	14.9%
ROW	8.4%
US	15.7%

Asset Weights	
PE	3.7%
RE	29.3%
NB	42.6%
EQ	3.9%
HY	2.9%
IG	15.9%
INFRA	1.8%

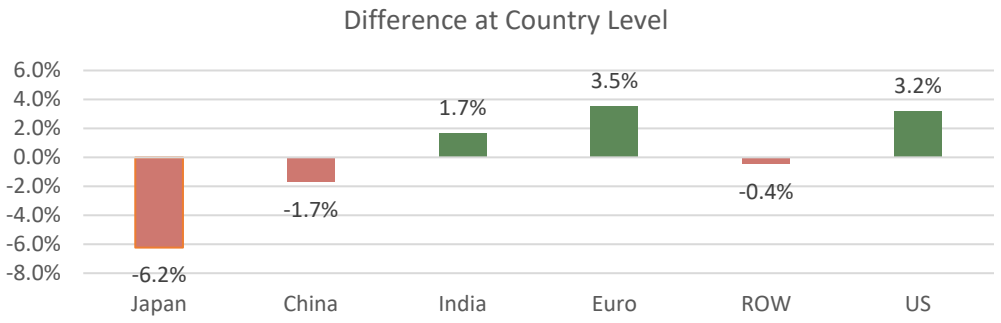


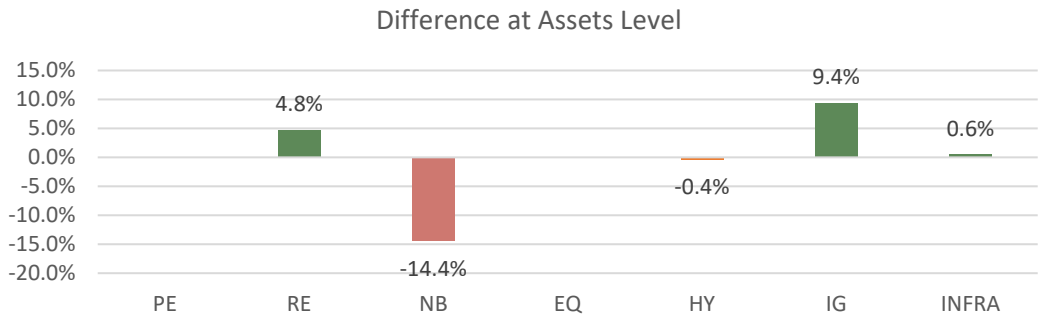
The weights concentrated in NB as its volatility is much lower than EQ

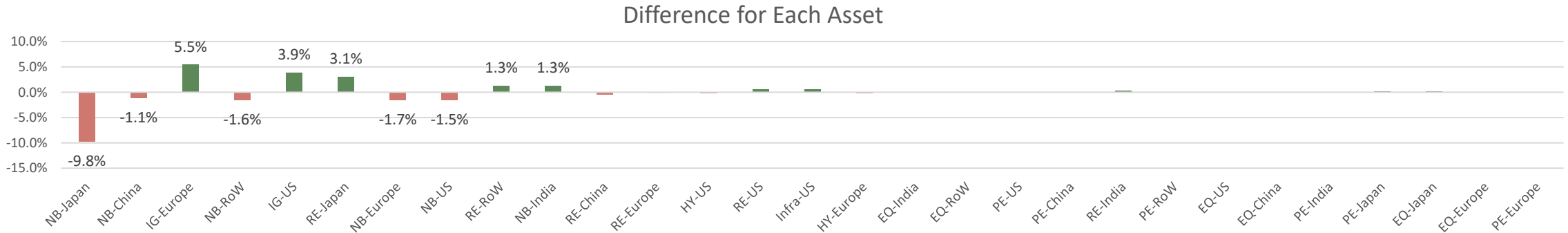


Two-level Hierarchical v.s. Multi-level Hierarchical

- NB Japan, itself as a cluster, has much lower volatility and less similarity with other assets
- In 2-Levels Hierarchical, NB Japan will be allocated much more weights due to its volatility and dissimilarity
- However, in Multi-Levels hierarchical, NB Japan will merge with NB in other countries first, which will increase cluster's volatility, and leads to lower weights allocation.







Conclusion

Summary:

- Hierarchical Clustering based strategy build more diversified and robust portfolios compare to classical methods, like Equal Weight and Risk Parity
- It takes the hierarchical structure into consideration and solves the stability problem compared to classical methods
- This strategy provides flexibility in choosing different allocation methods within or cross clusters, and the use of normal or tail correlation

Limitations:

- It is more difficult to setup the constraints at cluster level. And the portfolio does not generate the highest return compared to others
- It is best for deciding an anchoring point for unconstrainted portfolio, however the results may deviate from risk diversification under constraints
- It is purely based on historical risk assumption and relation, which may not still hold in the forward regime.
- The hierarchical tree will change with the risk regime. And the change of tree structure may lead substantial changes in the portfolio

Clustering-based Optimization

Methods	Details	Drawbacks
Hierarchical Clustering Asset Allocation (HCAA)	The principle is to find a diversified weighting by distributing capital equally among each cluster hierarchy.	portfolios may be heavily concentrated in terms of risk allocation
Hierarchical Risk Parity (HRP)	Aims at diversifying risk allocation.	Clusters can be too spread out, and not compact enough, and shape of dendrogram is not considered when the bisection is done.
Hierarchy Equal Risk Contribution (HERC)	HERC merges and enhances the machine learning approach of HCAA and the Top-Down recursive bisection of HRP, asset based on dengrogram and equal risk contribution allocation	

Course Assignment

- Download monthly returns data (since 2000) for US equities, US 10Yr treasury bonds, US REITs and US commodities
- Use full historical return/covariance assumptions to setup portfolio optimization with following objectives (no short constraint):
  - Maximize return subject to annualized volatility less than 10%
  - Minimize volatility subject to annualized return more than 4%
  - Maximize Sharpe Ratio
- If assuming all assets have the same Sharpe Ratio, what are the optimization portfolios with these three objectives? Will the optimization results depend on the Sharpe Ratio assumption?

Course Assignment

- Is there any regime change for the risk assumption observed from the historical data? If yes, what is the impact on the portfolio optimization results?
- Using the historical data, setup backtesting study for three risk-based optimal portfolios (equal weight, min vol, max diversification) for comparison and recommendation